

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-1986

Southwick 138 kV
March 01, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-1986. The Interconnection Customer proposes a 90.5 MW solar photovoltaic facility interconnecting at the Southwick 138 kV substation in the Virginia area. The request was received on January 16, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 441 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$9,021,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Dps Ocean Spray – Windsor Hills 138kV	Short Pump – S Meadow 2 138kV	125.5	230	14.9
Windsor Hills – Southwick Substation 138kV	Short Pump – South Meadow 2 138kV	124.3	260	14.9
Short Pump – S Meadow 2 138kV L	Southwick – Windsor Hills 138kV line	110.2	359	9.8
Short Pump – Mesa 138kV Line	Southwick – Windsor Hills 138kV	108.8	288	9.8
Pipers Gap – Mesa 138kV	Southwick – Windsor Hills 138kV	108.0	257	9.8
Pipers Gap – Dps Ocean Spray 138kV L	Southwick – Windsor Hills 138kV	107.0	229	9.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dps Ocean Spray – Windsor Hills 138kV	Rebuild line with higher-capacity conductor	\$7,576,000	42% (\$3,165,000)
Windsor Hills Substation – Southwick 138kV	Rebuild line with higher-capacity conductor	\$9,616,000	42% (\$4,018,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Short Pump – S Meadow 2 138kV L	Rebuild line with higher-capacity conductor	\$3,017,000	30% (\$897,000)
Short Pump – Mesa 138kV Line	Reconductor line	\$1,071,000	30% (\$318,000)
Pipers Gap Substation – Mesa 138kV	Reconductor line	\$1,005,000	30% (\$299,000)
Pipers Gap – Dps Ocean Spray 138kV L	Reconductor line	\$1,089,000	30% (\$324,000)
	Total allocated to project		\$9,021,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.